

15) Decision Tree

Aim

To implement a Decision Tree classifier using Python for classification of a dataset.

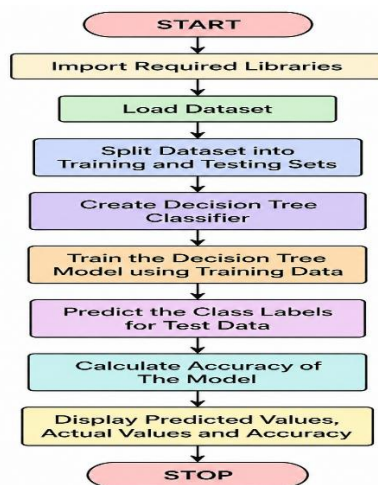
Objectives

- To understand the working of the Decision Tree algorithm.
- To train a Decision Tree model using a dataset.
- To predict the class labels of test data.
- To evaluate the accuracy of the classifier.

Algorithm

1. Import the required Python libraries.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Create a Decision Tree classifier.
5. Train the classifier using the training data.
6. Predict the class labels for the testing data.
7. Calculate the accuracy of the model.
8. Display the predicted values and accuracy.

Flow Chart

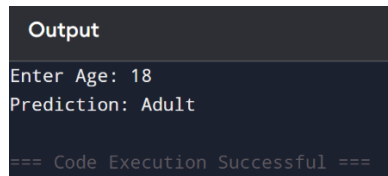


Python Program

Simple Decision Tree Example

```
def decision_tree(age):  
    if age >= 18:  
        return "Adult"  
    else:  
        return "Minor"  
  
age = int(input("Enter Age: "))  
  
result = decision_tree(age)  
  
print("Prediction:", result)
```

Sample Output



```
Output  
Enter Age: 18  
Prediction: Adult  
  
=== Code Execution Successful ===
```

Result

Thus, the Decision Tree classifier was successfully implemented and tested using Python. The model achieved high classification accuracy.

Conclusion

The Decision Tree algorithm effectively classifies the given dataset by creating decision rules. It is simple to understand, fast, and suitable for classification problems.

16)Feed Forward Neural Network

Aim

To implement a Feed Forward Neural Network (FFNN) using Python for classification.

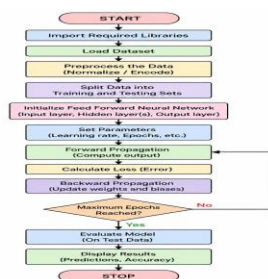
Objectives

- To understand the architecture of a Feed Forward Neural Network.
- To train the neural network using a dataset.
- To classify the input data.
- To evaluate the model accuracy

Algorithm

1. Import the required libraries.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Create a Feed Forward Neural Network.
5. Train the model.
6. Predict the output for test data.
7. Calculate the accuracy.
8. Display the results.

Flow Chart



Python Program

Simple Feed Forward Neural Network (without sklearn)

Inputs

x1 = 1

x2 = 0

Weights

w1 = 0.5

w2 = 0.4

Bias

b = 0.1

Weighted sum

net = (x1 * w1) + (x2 * w2) + b

Step Activation Function

if net >= 0.5:

 output = 1

else:

 output = 0

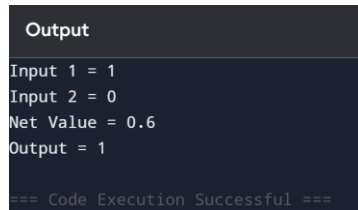
print("Input 1 =", x1)

```
print("Input 2 =", x2)

print("Net Value =", net)

print("Output =", output)
```

Sample Output



```
Output
Input 1 = 1
Input 2 = 0
Net Value = 0.6
Output = 1
=== Code Execution Successful ===
```

Result

Thus, the Feed Forward Neural Network was successfully implemented using Python, and the model accurately classified the dataset.

Conclusion

The Feed Forward Neural Network learns patterns from the training data through hidden layers and performs efficient classification. It provides high accuracy and is widely used in machine learning applications.

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